

HIGH & MIGHTY: Left, The *Vikrant*'s integrated ski jump ramp is a distinctive feature; below, Commodore Vidyadhar Harke, CO of *Vikrant*; bottom, Madhu Nair, chairman and MD, Cochin Shipyard, on *Vikrant*'s deck





WHEN RUSSIA WOULD NOT SUPPLY STEEL, DMRL, A PART OF DRDO, DEVELOPED WARSHIP-GRADE STEEL, A MAJOR ACHIEVEMENT

steel, DMR 249, cheaply, and produced tens of thousands of tonnes of it. It ended the nation's decades-old reliance on import for military-grade steel," Nair tells INDIA TODAY. Based on the manufacturing experience of IAC *Vikrant*, eight anti-submarine shallow watercraft corvettes and six next-generation missile vessels are currently under construction at CSL.

Anjana K.R., head of the design division of the CSL, says that they worked with multiple global stakeholders during the making of the *Vikrant*. The warship is designed jointly by the CSL and the design bureau of the Indian Navy, which used several design iterations (a cyclical process involving testing, analysing and refining), including the use of 3D virtual reality models and advanced engineering software. "*Vikrant*'s propulsion system integration is Italian-made, the aircraft facility complex, including arresting gears, STOBARs (short take-off but arrested recovery, the system for launch and recovery of aircraft) and flight handling equipment is from Russia, the aircraft lifts are from the UK, the ammunition lift is from the US and its hangar doors are Swedish," says Anjana. However, 76 per cent of the equipment is indigenously sourced. On design challenges, Anjana remembers the flight deck. The IAC *Vikrant* was initially designed like its eponymous ancestor—which had a separate, not integrated, ski jump (the upward curved ramp that assists a plane to take off) with a flat deck. "CSL had built three-fourths of the warship when it changed the layout of the flight deck. We had reached up to the front portion of the ship, when the navy came up with a suggestion to change the shape," Anjana says, adding that *Vikrant* has gone through over 5,000 changes from its original plan.

IAC *Vikrant* has completed 25 days of sea trials since August 2021, followed by second and third phases of trials in October 2021 and January 2022, respectively. Madhu Nair says that during its first sortie, when it was out to sea for six days, the ship managed to achieve full speed—a rare achievement in the trials phase. Also, 1,600 men were on board in all sea trials, during which the navy carried out integrated checks on equipment and systems, including the critical aviation facilities equipment that controls the operation of the carrier-based aircraft.

Several world powers are now operating or building technologically advanced aircraft carriers to safeguard their maritime rights and interests. Indeed, aircraft carriers are the flagships of most powerful navies, with the US leading the way. There are a total of 41 active carriers operated by 13 navies across the world. The US, UK, Russia, Italy and France are foremost among nations that operate carriers. India's biggest adversary in IOR, the People's Liberation Army-Navy (PLA-N), aims to have a fleet of over 10 aircraft carriers by 2050. Its third